α-labeling of banana tree graphs

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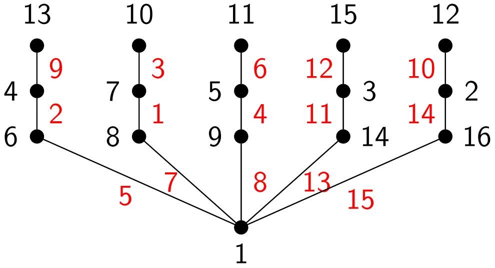
Sections

Motivation

- Fokus rada: α-**označavanje** i posebna familija stabala – banana stabla.
- Rosa (1967) uveo graciozno (β) označavanje i α-označavanje.
- Ringel-Kotzigova hipoteza: Sva stabla su graciozna.
- Primjene označavanja stabala: teorija kodiranja, teorija mreža.

Graceful labeling

- $f : V \to \{1, 2, ..., m + 1\}$
- $f^*(uv) = |f(u) - f(v)|, \; uv \in E$
- f^* must be a bijection to $\{1, ..., n - 1\}$

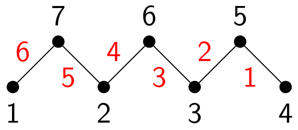


Graceful labeling of a tree

α-labeling

- Additional property: there exists α such that for each edge uv exactly one inequality holds:
  1. $f(u) \leq \alpha < f(v)$
  2. $f(v) \leq \alpha < f(u)$

TODO: Mozda dodati kompleksnije stablo



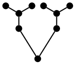
α-labeling of a tree

Conjugate and reverse α-labelings

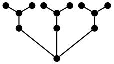
TODO: dodati Rosin remark o α-označavanjima

Definition of a banana tree $B(n, k)$

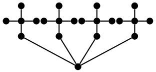
- Hierarchical structure: n star graphs S_k
- Each star is connected to a central vertex (root)



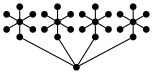
$B(2, 4)$



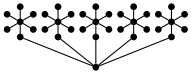
$B(3, 4)$



$B(4, 5)$



$B(4, 7)$



$B(5, 7)$

Symmetries of banana trees

TODO: dodati simetrije banana stabla

The set of labelings $B(n, k)$

Definition

Let $\mathcal{A}(n, k)$ be the set of all α-labelings of the $B(n, k)$ tree and $\mathcal{A}_x(n, k)$ be the set of all α-labelings of the $B(n, k)$ tree where $f(r) = x$. Let the set of principal α-labelings $\mathcal{P}(n, k)$ be defined as $\mathcal{P}(n, k) = \bigcup_{x=1}^{\lfloor \frac{n}{2} \rfloor + 1} \mathcal{A}_x(n, k)$. Lastly, let the set of *central* α-labelings $\mathcal{C}(n, k)$ be defined as $\mathcal{C}(n, k) = \mathcal{A}_{\lfloor \frac{n}{2} \rfloor + 1}(n, k)$.

Proposition

Let f be an α-labeling of the $B(n, k)$ tree for any $n \in \mathbb{N}$ and $k \geq 2$. Then,

$$f(r) \leq \alpha \iff f(c_i) \leq \alpha \quad \forall i \in [n]$$

Partitioning $\mathcal{A}(n, k)$

Theorem

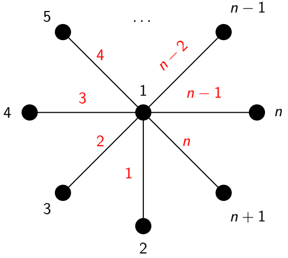
For every $n, k \in \mathbb{N}$, we have

$$\mathcal{A}(n,k) = \mathcal{P}(n,k) \cup \mathcal{P}^r(n,k) \cup \overline{\mathcal{P}}(n,k) \cup \overline{\mathcal{P}^r}(n,k),$$

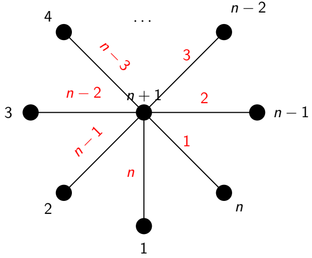
where $\mathcal{P}^r(n,k) = \{f^r : f \in \mathcal{P}(n,k)\}$, $\overline{\mathcal{P}}(n,k) = \{\overline{f} : f \in \mathcal{P}(n,k)\}$ and $\overline{\mathcal{P}^r}(n,k) = \{\overline{f^r} : f \in \mathcal{P}(n,k)\}$.

- Reducing the set of possible α-labelings by a factor of 4
- Reducing the complexity of the proofs

Classifying all the α-labelings of $B(n,1)$ and $B(1,k)$ trees



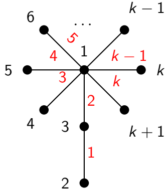
α-labeling f_1 of $B(n,1)$



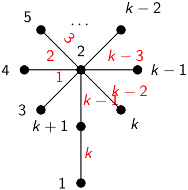
α-labeling f_2 of $B(n,1)$

Figure: Two α-labelings of $B(n,1)$

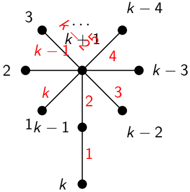
Classifying all the α-labelings of $B(n, 1)$ and $B(1, k)$ trees



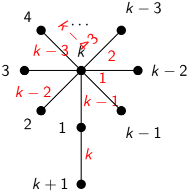
α-labeling f_1 with $\alpha_1 = 2$



α-labeling f_2 with $\alpha_2 = 2$

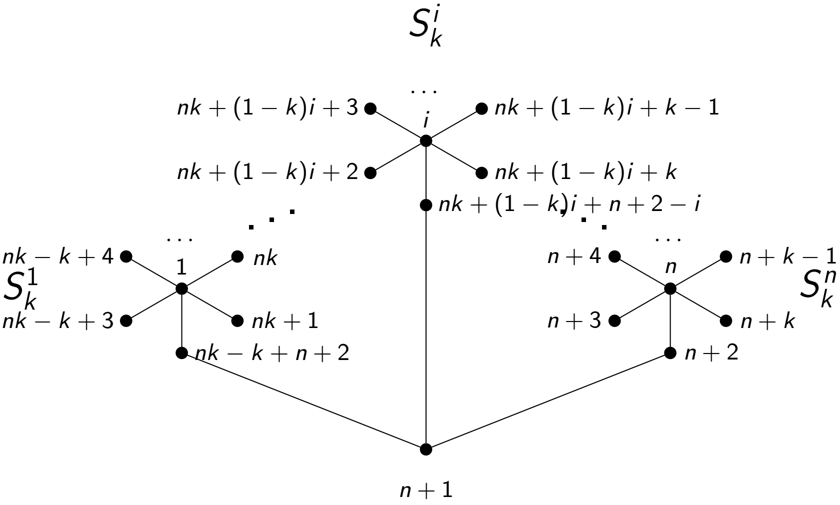


α-labeling f_3 with $\alpha_3 = k - 1$



α-labeling f_4 with $\alpha_4 = k - 1$

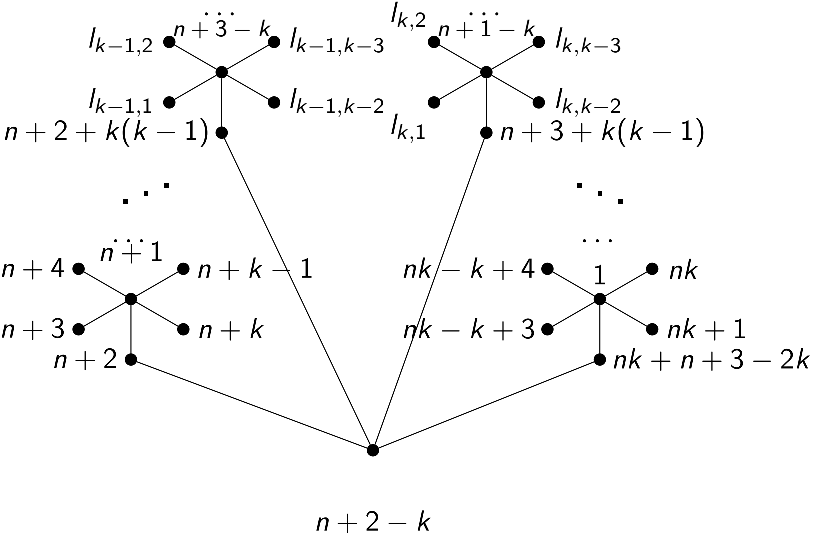
A sketch of the proof for $B(n, k)$ for $k \geq n$



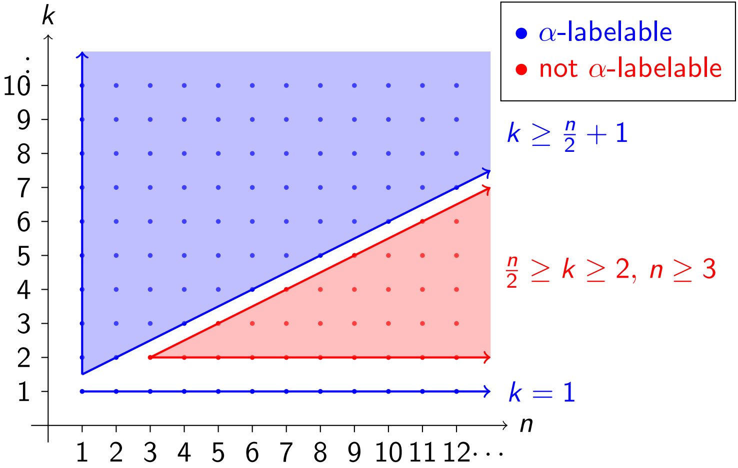
A sketch of the proof for $B(n,k)$ for $\frac{n}{2}+1 \leq k \leq n$

$$
\begin{aligned}
f(r) &= n+2-k \\
f(c_i) &= \begin{cases} n+2-i, & i<k \\ n+1-i, & i \geq k \end{cases} \\
f(a_i) &= \begin{cases} n+2-k+ki, & i<k \\ n+3-2k+ki, & i \geq k \end{cases} \\
\mathcal{F}(\mathcal{L}_i) &= \{n+3-k+(k-1)i, \ldots, n+1+(k-1)i\} \setminus \{f(a_i)\}
\end{aligned} \tag{1}
$$

A sketch of the proof for $B(n,k)$ for $\frac{n}{2}+1 \leq k \leq n$



α-labelability of $B(n,k)$ trees



Conclusion

- We settled the open question of the existence of α-labelings for $B(n, k)$ trees for all $n, k \in \mathbb{N}$
- Introduced new notation and concepts for α-labelings of $B(n, k)$ trees
- Opened new questions regarding the number of non-isomorphic α-labelings of $B(n, k)$ trees for $n \geq 3$

Thank you for your attention!

Any questions?